Week 8 Slides — Speaker Script

Independent Research Project II — Execute · Stress-Test · Write · Defend

NGN Empirical Finance Research Camp · Prof. Lei Gao (GMU) · Friday, August 14, 2026

Full speaker notes for every slide (also embedded in the .pptx Notes pane). 24 slides.

Slide 1 — Title — Independent Research Project II

Welcome to Week 8 — the last live session, and the one every other week was built for. Before we start, drop your Week 7 deliverable in the chat: your filed pre-analysis plan and identification memo. If you did the homework, you also arrive with a dataset in hand. Good. Here is what today is. You filed the plan; now you execute it and you defend it. In the next two hours you will run your one pre-registered test, try to break your own result on purpose, write it up to publication standard, and defend it out loud to the room. And the through-line of the whole camp finally lands: a result you would defend to a skeptic. Whatever comes back — a big effect or a flat null — you report it honestly and you defend the design. The capstone paper is due tomorrow, August 15. This session is your last supervised push before it. Let's cross the finish line.

Slide 2 — The 8-week map — where we are

Find yourself on the map one last time. Weeks one through four built the toolkit — probability, the OLS engine, two full weeks of causal design. Weeks five and six taught you to read the frontier, by hand and then with AI. Week seven was the pivot: you turned a question into a filed pre-analysis plan. And here we are — Week eight, today, Research Project Two, where you execute that exact plan and defend what it produces. Notice the tagline on the slide: you filed the plan, now you execute and defend it. There is no week nine. This is the last supervised session before your capstone is due, so everything today is real work on your real project, not a demo on mine. The skills weeks are long behind you. From here on out you are the researcher, and by two o'clock you will have run, stress-tested, written, and defended a study that is entirely yours.

Slide 3 — The arc of the day — five moves

Here is the whole day in one arc — five moves, in order, and none of them is optional. First, execute: you run the one primary specification you pre-registered, and you report it first, whatever it says. Second, break it on purpose: you attack your own result with tests that are genuinely capable of failing, because a result that survives an honest attack is one people believe. Third, write: you turn the run into a paper — a disciplined abstract, a clean contribution, causal language matched to your design. Fourth, defend: you present it in eight minutes and take the room's hardest question without flinching. And fifth, submit: tomorrow the capstone goes in. Notice the shape. You do not get to skip from execute straight to write. The breaking-it step in the middle is where honest research lives. Most students want to run it and love it. Today you run it and try to kill it. That is the difference.

Slide 4 — Block A — Execute & Stress-Test

Block A — execute and stress-test. This is the half of the day people find hardest emotionally, because it asks you to attack the result you have been hoping for. Here is the sequence. First, the confirmatory run: your one pre-registered spec, reported first, no matter what. Then the specification curve — the realization that your one spec is a single point in a cloud of hundreds of defensible choices, and we are going to plot the whole cloud at once. Then reading that curve: telling a robust result from a fragile one. And then the full robustness battery — alternative standard errors, placebo tests, a bootstrap built for few clusters, and a sensitivity check for unobserved selection. The goal of this whole block is not to protect your result. It is to find out whether it deserves to survive. Let's run it and then try to break it.

Slide 5 — The confirmatory run — report it first

Start with the confirmatory run, and treat this as a discipline, not a formality. You wrote down one primary specification last week, before you saw any outcome. Today you run exactly that specification, and you report it first — before any robustness check, before any alternative, whatever the number turns out to be. If it is a clean significant effect, wonderful. If it is a flat null, you report the null with the same steadiness, because you pre-committed and a null from a pre-registered test is real evidence, not a failure. Now, real research is messy, and you will sometimes have to depart from the plan — a variable was unavailable, a sample restriction did not survive contact with the data. That is fine, on one condition: every departure goes into a deviations log. You do not quietly erase the change; you write it down, in the open. Transparency, not erasure. The log is what keeps an honest deviation from looking like a fishing expedition.

Slide 6 — The specification curve — the multiverse

Now the idea that will change how you read every empirical paper for the rest of your life: the specification curve. Your primary spec made a whole series of choices — which controls to include, which sample window, which fixed effects, which estimator. Each of those choices was defensible, and so were the alternatives you did not pick. Multiply them together and there are not one or two reasonable specifications — there are hundreds. That whole set is the multiverse. Instead of quietly running the one that looks best, you run all of them and plot every estimate at once, sorted from most negative to most positive. This is specification curve analysis — Simonsohn, Simmons and Nelson, 2020, in Nature Human Behaviour. The picture on the slide is simulated, illustrative only — not a real estimate. But the discipline is real: you stop cherry-picking a specification and start showing the reader the entire cloud your one headline number lives inside.

Slide 7 — Reading the curve — robust vs fragile

So how do you read a specification curve? Two words: robust versus fragile. A robust result holds its sign and its significance across most of the curve. Toggle a control on or off, swap the sample window, change the estimator — the headline barely moves. That is a finding people believe. A fragile result is the opposite: it lives on a few lucky specifications and vanishes everywhere else. Change one reasonable choice and the effect collapses or flips sign. That is not a discovery; that is a coincidence you got attached to. Here is the honest move, and it is the whole point: you report where on the curve your headline specification sits. Is it typical of the cloud, or is it one of the few outliers holding the story up? The simulated curve on the slide shows both cases side by side. Telling the truth about where you sit is what separates a researcher from a salesperson.

Slide 8 — The robustness battery — tests built to fail

The robustness battery — four tests, each designed to be capable of failing, because a test that cannot fail proves nothing. One: alternative standard errors. Re-cluster, try heteroskedasticity-robust errors, and see whether your significance survives a different error structure. Two: placebo and falsification tests — the moves from last week — a fake treatment date, an unaffected group, checks that should come back empty if your design is clean. Three: the wild-cluster bootstrap. When you have only a handful of clusters — a few states, a few industries — ordinary clustered standard errors lie to you, and the wild-cluster bootstrap gives you honest p-values in that small-cluster world. Four: an Oster delta check — a sensitivity analysis that asks how much unobserved selection, relative to what you already control for, it would take to explain your whole result away. And if you test many outcomes, apply a Benjamini-Hochberg FDR correction so you are not celebrating one lucky star out of twenty. Survive these, and people believe you.

Slide 9 — Capable of failing — the whole idea

Let me put the philosophy of the whole block on one slide, because it is the single most important idea in empirical research. A result that survives tests genuinely capable of failing is one people believe. Read that carefully — capable of failing. If you only run checks that were always going to pass, you have proven nothing; you have performed a ritual. The value of a placebo test is precisely that it could have lit up and exposed you. The value of a specification curve is that your effect could have vanished across the cloud. The value of an Oster check is that a small amount of selection could have explained you away. When you run tests that had a real chance of killing your result, and the result is still standing, you have earned the reader's belief. Fragile findings dodge those tests. Robust findings walk straight into them. Be the second kind of researcher.

Slide 10 — Causal-language discipline — match verb to design

Now the single most common capstone mistake, and I want you to catch it in your own draft before I do: causal language that your design has not earned. Match the verb to the design. If you ran an observational study — a regression on data as it came, no clean shock, no experiment — then your result is that X is associated with Y. You write associated with, correlated with, predicts. You do not write causes, raises, lowers, drives. Those causal verbs are only yours when your identification earns them — a clean natural experiment, a real quasi-random shock, the kind of design we spent two weeks building. Here is why it matters so much: over-claiming is the fastest way to lose a referee. The moment a skeptical reader sees a causal verb sitting on top of a correlational design, they stop trusting the entire paper. Calibrated language is not weakness. It is the mark of a researcher who knows exactly what their design can and cannot say.

Slide 11 — Block B — Write It Up

Block B — write it up. You have a result and you have stress-tested it. Now you turn it into a paper that meets a real publication standard, and I am going to give you the machinery to do it fast. First, the skeleton the whole paper hangs on — five sentences and a disciplined abstract. Then the one sentence that says what is new. Then the causal-language discipline we just covered, applied to your own prose. And finally, the tables and figures at publication quality plus the one-click replication packet — the repository anyone can rerun with a single command to regenerate every number in your paper. Writing is not the thing you do after the research; the writing is where you find out whether you actually understand your own result. Let's build the paper.

Slide 12 — The empirical-paper structure — six parts

Every empirical paper has the same skeleton, and once you see it you can never unsee it. Six parts. The abstract: your whole paper in 250 words. The introduction: the question, why it matters, and what you find — front-loaded, because a referee decides in the first page whether to keep reading. Data and design: where the data came from and the identification strategy that makes your estimate mean something. Results: the primary specification first, then the robustness battery. Discussion and limitations: what it means, and — just as important — what it does not mean and where the design is vulnerable. And the piece that is unique to honest empirical work: a deviations log, the transparent record of every departure from your pre-analysis plan. Notice that limitations and the deviations log are not afterthoughts you bury. In a credible paper they are load-bearing. Stating your own weaknesses first is how you earn the reader's trust in your strengths.

Slide 13 — Five-sentence skeleton + the 250-word abstract

Here is how you write the paper fast: get the skeleton right before you write a single paragraph. Five sentences. One — the question: what do you ask? Two — the design: how do you identify the answer? Three — the data: what do you use to do it? Four — the finding: what did you actually get? Five — the contribution: why does anyone care? If you can write those five sentences cleanly, you can write the whole paper, because everything else is just expansion. Then the abstract, and here the constraint is the teacher: 250 words, hard budget. That limit forces you to cut everything that is not the question, the design, the data, the finding, and the contribution. A sprawling abstract is a sign of unclear thinking; a tight one proves you know what your paper is about. Write the five sentences, then grow them into 250 words. Not the other way around.

Slide 14 — The one-sentence contribution

Now the hardest sentence in the paper, and the one referees read most carefully: the one-sentence contribution. It fills in a single blank — we are the first to show that blank. And that sentence carries two burdens at once. It must be true: if three papers already showed it, you are not first, and a referee who knows the literature will catch you instantly. And it must be supported: your own evidence has to actually back the claim you just made. The most common failure is a contribution sentence that reaches past the results — claiming the paper shows something bigger or more causal than the design can support. So write it, then interrogate it twice. Is it genuinely new? And does my own table on the next page actually prove it? When the answer to both is yes, you have a contribution. When either is no, you shrink the sentence until it is true and supported. A smaller claim you can defend beats a grand one you cannot.

Slide 15 — Tables & figures + the one-click replication packet

Two craft standards to close the writing block. First, tables and figures to publication quality. A good table stands on its own — a reader should understand it without the text: clear variable names, standard errors in parentheses, significance marked, units and sample stated in the notes. A figure should make one point cleanly. If your reader has to hunt for what a table is telling them, it is not done. Second, and this is the one that separates a real study from a nice-looking one: the one-click replication packet. That is a repository — code, data or a documented data-pull, and a single command — that anyone, including future-you, can run to regenerate every single number in your paper from scratch. Not most numbers. Every number. If you cannot rerun your whole paper with one command, it is not reproducible, and reproducible is the standard the whole field is moving toward. Build the packet as you go, not the night before.

Slide 16 — Block C — Defend It

Block C — defend it. This is the payoff of the whole camp, so we protect the time. You have a result, you have stress-tested it, and you have written it up. Now you have to stand in front of a skeptical room and defend it out loud. Three things in this block. First, peer review from the other side of the table — reading your own claims the way a referee would, and knowing what a referee actually asks. Second, the eight-minute talk, decomposed minute by minute, because a defense is a performance with a structure. And third, the three honest answers that carry you through any Q and A. This is where you learn that your result is yours to defend — not mine, not a textbook's. Yours. Let's get you ready to stand behind it.

Slide 17 — Peer review — the referee report & the R&R memo

Peer review — and the fastest way to survive a referee is to become one first. Two documents live here. The referee report: a structured critique of a paper — what is the contribution, is the identification credible, is the causal language calibrated, what is the one fatal question. And the revise-and-resubmit memo: the author's point-by-point response, where you either fix the problem or explain, respectfully, why the referee is wrong. Today you read your own paper through the referee lens. Look at your headline claim and ask the hard question a referee would ask. Is my causal language calibrated, over-claimed, or under-claimed? What is the single objection that could sink this? Our mentor anchor is exactly this discipline: Elnahas, Gao, Hossain and Kim, published in the Journal of Financial and Quantitative Analysis in 2024 — a paper we read specifically to learn what a referee actually asks. When you can referee yourself honestly, an outside referee holds far fewer surprises.

Slide 18 — The 8-minute talk, decomposed

The eight-minute talk, decomposed — because a defense is not a data dump, it is a structure. Here is the budget. One minute of motivation: why should anyone care about this question? Two minutes of design: how do you identify the answer, and why is it credible? Two minutes of results: your primary specification, stated cleanly. One minute of robustness: how you tried to break it and how it held. One minute of contribution: the one new thing. And one minute of limitations: what the design cannot say. Eight minutes, and the governing rule is one idea per slide. Do not cram. A slide with one clear idea lands; a slide with five ideas lands none of them. Notice that a full minute — an eighth of your talk — goes to limitations. That is not modesty; it is strength. The presenter who names their own weaknesses looks like the smartest, most trustworthy person in the room. Rehearse it to time.

Slide 19 — The three honest answers in Q&A

Q and A terrifies people, so let me hand you three honest answers that carry you through almost anything. Answer one, when you genuinely do not know: I don't know — and here is exactly how I would find out. That is a strong answer, not a weak one; it shows you think like a researcher. Answer two, when someone hits a real weakness: that is a limitation, and here is how I bounded it — here is how big the problem could be and why it does not overturn my result. Answer three, when someone doubts your finding: here is the exact evidence, and you point to the specific table or figure. Notice what all three refuse to do: bluff. The one fatal move in a defense is to make something up to look confident, because the moment a sharp questioner catches it, you lose the whole room. Honesty, precision, and knowing where your own evidence lives — that is what wins a defense.

Slide 20 — Submission & the long arc

Step back from tomorrow's deadline and see the long arc, because today is a beginning, not an end. Your capstone is a poster and a paper, but that is step one of a real road. There is the Young Scholars Journal, where strong high-school and early-undergraduate research actually gets published. There is posting a preprint to SSRN, so your work has a permanent, citable home the day it is done. There is the conference circuit — presenting, getting feedback, meeting people who do this for a living. And there is the decade arc: a line of researchers whose story starts with a poster exactly like the one you are building and, years later, ends in a top journal. I am not promising you that path — I am telling you it is real and it starts here. The habits you are practicing today — pre-register, stress-test, calibrate, replicate — are the same habits that carry a paper all the way to the top. Today's poster is the first step.

Slide 21 — Workshop — the mock defense

Workshop — the mock defense, and this is the single most valuable exercise of the day, so we run it live. Here is the drill. Each of you, in turn, does two things. First, you give your one-sentence contribution out loud — we are the first to show that blank — clean, true, and supported. Then you take one referee question from the room, and you answer it using one of the three honest answers we just practiced. That is it. Short, sharp, real. The rest of the room's job is to be the referee: listen to the contribution and ask the one hard question — is that causal claim earned? Is that really new? Where is the evidence? Catching a weak spot here, in this room, today, is free. Catching it tomorrow when you are standing in front of the actual audience is not. So when it is your turn, say it clearly, take the hit, and answer honestly. Let's go around.

Slide 22 — Your capstone checklist — four gates

Before you leave, here is your capstone checklist — four things, and the submission is not complete until all four exist. One: the paper itself — abstract, intro, data and design, results, discussion and limits, deviations log. Two: the one-click replication packet — the repository that regenerates every number with a single command. Three: the eight-minute presentation — decomposed to time, one idea per slide, rehearsed. And four: the AI-use disclosure — an honest statement of where and how you used AI in the work, because the same transparency we demand of the data we demand of the tools. Miss any one of these and the submission is incomplete. This is the terminal deliverable of the entire program, so treat the checklist as a gate, not a suggestion. Print it, and do not submit until every box is checked.

Slide 23 — Wrap & final assignment — the Capstone Submission

Let's wrap and set the final assignment. This is it — the Capstone Submission, the terminal deliverable of the whole camp. Four pieces: the completed paper, the one-click replication packet, the eight-minute presentation, and your AI-use disclosure. It is due tomorrow, August 15 — presentation and conference day. Everything from the last eight weeks converges on this one submission. So tonight you finish the run if it is not finished, you make sure the packet actually reruns clean on a fresh machine, you rehearse the talk to eight minutes, and you write the honest AI disclosure. Then you submit. I want to say this plainly: what you are handing in is a real piece of empirical research that you designed, executed, stress-tested, wrote, and defended yourselves. Eight weeks ago most of you had never run a regression. Tomorrow you present a study you would defend to a skeptic. Go finish it. I could not be prouder of this group.

Slide 24 — Thank you & references

Let me close with the line that has run through all eight weeks. A result you would defend to a skeptic — that has been the standard from day one, and today you met it. You executed a pre-registered test, you tried to break your own result on purpose, you wrote it up with language matched to your design, and you defended it out loud. Whether your result came back a big effect or a flat null, you reported it honestly, and that is what makes you a researcher. The verified anchors behind today: Simonsohn, Simmons and Nelson, 2020, in Nature Human Behaviour, on specification curve analysis; Olken, 2015, in the Journal of Economic Perspectives, on pre-analysis plans; the mentor anchor Elnahas, Gao, Hossain and Kim, 2024, in the Journal of Financial and Quantitative Analysis, for what a referee actually asks; and the running design anchors from Prof. Gao's work — the Rainbow of Municipal Credit, Gao and Sun's 2019 PNAS fair-lending study, and Deng, Gao and Kim's 2020 Reg SHO paper in the Journal of Corporate Finance. Your capstone is due tomorrow, August 15. Go plant your flag. It has been an honor.